

CONCENTRATION, NOT UNCERTAINTY: WHY TARGETED SYNTHETIC DATA DOESN'T HELP CAMOUFLAGED OBJECT DETECTION

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ABSTRACT

Camouflaged object detection needs pixel-accurate masks that are slow and costly to draw, so synthetic training images are a standard remedy. That raises a question nobody has answered: under a fixed budget, which images should you generate? The obvious answer is to let the model choose, and we built that loop — cluster the unlabelled real images the model will face, find the clusters it is least sure about, aim the generation there, retrain, repeat.

No advantage survives measurement. Across 103 trained runs, targeting does not beat random allocation, and five independent controls each close a different reason that null might be an artifact. The reason is itself a positive finding: a targeted training set *is* measurably different from a random one, but the difference comes from concentrating the budget rather than aiming it — every rule that preserves the allocation's shape reproduces the effect, including one that reverses the uncertainty signal outright. Separately, 50 of CHAMELEON's 76 images are re-encoded copies of training data, while the standard check reports 0 overlap.

1 INTRODUCTION

Camouflaged object detection asks for a pixel-accurate outline of an animal that has evolved not to have one. Its training masks are unusually expensive for exactly that reason: an annotator must decide where a boundary lies when the point of the object is that its boundary is invisible. One standard response to the resulting scarcity is to manufacture more data — a generator takes a real cut-out object and paints a fresh background behind it, producing a new labelled pair at no annotation cost.

Generation is not free either, though, and every generated image has to be trained on. So if you can afford a thousand of them, which thousand? The obvious answer is to let the model choose: train it on what you have, look at the unlabelled real images it will actually be judged on, find the ones it handles least confidently, and aim the generation there. Every ingredient of that loop is standard — uncertainty sampling from active learning (Settles, 2009), a student–teacher disagreement score (Tarvainen & Valpola, 2017), clustering over self-supervised features (Oquab et al., 2024) so the budget spreads rather than piling onto near-duplicates. The composition is ours, and we could not make it work.

The reason turned out to be more interesting than the failure. A targeted training set really is measurably different from a randomly chosen one of the same size — train a probe to tell the two apart and it succeeds easily, on images it has never seen. But that difference does not come from the aiming. It comes from the fact that aiming *concentrates* the budget on a few clusters, and any rule that concentrates it the same way reproduces the effect: pointing the budget at the wrong clusters reproduces it, and so does reversing the uncertainty signal outright. What looked like the effect of aiming was the effect of concentrating, which means the control that settles the question is not a random baseline but a same-shape allocation with the signal destroyed inside it.

On trained accuracy, the direction of the signal moves nothing this design can resolve. That null is bounded rather than absolute — an interval on the two conditions actually compared excludes the improvement this approach reports, while admitting effects a few times smaller — and it is defended rather than asserted. A positive control confirms the measurement detects a real effect. Substituting

a perfect score for ours changes nothing. Removing the scheduling artifact that made added images displace existing ones changes nothing. And re-running the decisive comparison at eight seeds per condition leaves the gap *smaller* than it was at three.

Separately, and independently of whether our method worked, we found that the benchmark such claims are routinely checked against is mostly training data, and that the standard check for this reports it clean.

Contributions. The first two concern the method we built; the last two travel beyond it.

- **Concentration, not the uncertainty direction**, is what makes a targeted synthetic training set measurably different from a random one — demonstrated in feature geometry, in trained boundary accuracy, and across eight seeds of trained accuracy.
- **A bounded null on the direction**, defended by five independent controls — a positive control, a perfect-score oracle, an unpinned optimisation schedule, a same-shape shuffle and a seed expansion — rather than asserted once.
- **Three checks that cost no training** and should precede any cluster-wise acquisition policy: whether the target data clusters at all, what the funded clusters are made of, and how much of the endpoint’s error sits between clusters rather than within them.
- **CHAMELEON is 50 of 76 images (65.8%) training data** for models trained on COD10K, while exact hashing reports 0 overlap; we release a re-encode-tolerant detector, the confirmed pair list and a clean evaluation protocol.

1.1 RELATED WORK

This paper does not propose a detector. It asks whether an *acquisition policy* — a rule for deciding which synthetic images to generate under a fixed budget — can improve a pipeline that is otherwise held fixed. The task it does that on is camouflaged object detection, which asks for a pixel-accurate outline of an object whose colour and texture are built to defeat exactly that; Fan et al. (2023) survey its architectures, datasets and evaluation conventions.

What we measure, and what decides. A prediction is scored by how well its mask matches the true one. Our primary measure is the *S-measure* S_α (Fan et al., 2017), which rewards getting an object’s *shape* right rather than counting correct pixels, and which runs from 0 to 1 with higher better. The scale matters for reading everything that follows: every detector in this paper scores near 0.70, so the differences the paper turns on are small fractions of the range these models actually occupy. Three words then recur. An **arm** is one training condition; all arms share a base pool of labelled images and differ only in which images are appended to it. A **seed** is one training run of one arm — repeat runs of the same arm do not return the same score, and the size of that seed-to-seed spread is what any real difference has to beat. An **endpoint** is a held-out test set a verdict is read from. Finally, the **reference effect** is the improvement this repository reproduced for the published method it builds on, $0.7030 \rightarrow 0.7172$, or $+0.0142$. That is the size of effect this design had to be able to resolve, and the yardstick every sensitivity claim in this paper is measured against.

The systems this one sits on. The closest are LAKE-RED (Zhao et al., 2024), whose background generation and compositing we use unchanged, and S2R-COD (Luo et al., 2025), whose synthetic-to-real self-training protocol we follow. Relative to both, the only thing we add is an allocation layer that chooses *which* generated images to append; the generator, the budget and the training procedure are left alone. Synthetic-augmentation methods such as DatasetDM (Wu et al., 2023) address how to construct labelled data at all, which is a different question from how to spend a fixed amount of it.

Where the allocation rule comes from. Every part of it is standard. Uncertainty sampling comes from active learning (Settles, 2009), except that our query is answered by a generator rather than by a human annotator; the uncertainty is student–teacher disagreement of the kind mean-teacher training already minimises (Tarvainen & Valpola, 2017); and *k*-means over frozen self-supervised features (Oquab et al., 2024; Radford et al., 2021) keeps the budget from piling onto near-duplicates,

in the spirit of core-set selection (Sener & Savarese, 2018). Other constructions exist — deep ensembles (Lakshminarayanan et al., 2017), MC-dropout (Gal & Ghahramani, 2016), BADGE (Ash et al., 2020) — and Section 3.3 measures two of them alongside ours, but we propose none. What is contributed here is a controlled test of the composition, not a new estimator.

The control that is missing from this literature. Three recent systems share the premise that a generation budget should be steered rather than spent uniformly. GAUDA (Frisch et al., 2025) conditions a latent diffusion model on the classes a Bayesian model is least certain about; DisCL (Liang et al., 2025) steers image-guidance strength along a curriculum; SynQuE (Chen & Zhong, 2025) ranks whole synthetic datasets using only unannotated real data. **None of the three runs a control that holds the allocation’s shape fixed while destroying the signal inside it**, and that turns out to be the control that decides the question, because a steered selection differs from a random one in two ways at once — where the budget points, and how unevenly it is spread. Appendix A places these systems side by side and covers benchmark definitions and generative alternatives. Section 2 specifies the loop we built and the arms that separate those two things.

2 METHODS

The idea is easy to picture. Train a model on the labelled images you have; look at the unlabelled real images it will be judged on and find the ones it handles least confidently; ask a generator for new images resembling those, add them to the training set, and train again. What follows is our instantiation of that idea, specified exactly, because a null result is only as informative as the thing it tested.

The setting. We train on a *source pool* of 4447 labelled image and mask pairs and want the model to work on a *target pool* of 4040 unlabelled real images, with verdicts read from two held-out endpoints: COD10K-test decides and NC4K is reported. The target pool is not one distribution but a mixture of 3040 COD10K images and 1000 from CAMO, which matters later (Section 3.3). Training runs in two rounds: a student trains alongside a teacher that is a slowly updated running average of its own weights, target images the two already agree on are appended between rounds with the teacher’s prediction as their label, and round two retrains from scratch on the enlarged pool. Two properties the phrase “closed loop” hides both bound what follows — the round count is a fixed constant, not a convergence criterion, and the generator renders its pool once, offline, and is never invoked during training.

The allocation, in four stages. A fixed generation budget has to be spent somewhere, and our rule spends it where the model is weakest. **(i) Group the target images**, so that “weak” can name a region of the data rather than a single image: k -means into $k = 75$ clusters over L2-normalised self-supervised embeddings, fitted once and never refitted. **(ii) Score each image for uncertainty.** We need a number, computable without labels, that says how unsure the model is; the natural candidate is how far the student and its own running average disagree, which is available precisely because the consistency loss is already trying to reduce it:

$$ES(x) = a \|\nabla\sigma(s(x)) - \nabla\sigma(t(x))\|_1 + b \text{BCE}(\sigma(s(x)), \sigma(t(x))), \quad (1)$$

with student and teacher logits s, t , sigmoid σ , Sobel gradient magnitude ∇ , and configured weights a, b in Appendix B. What this buys is a score needing no ground truth, computable on exactly the images the model will be judged on; a cluster’s score is the mean over its members. Its dominant term is a distance between *gradient* magnitudes, which makes it a boundary-focused signal — a fact Section 3.2 returns to. **(iii) Turn scores into a budget split.** The budget is spread by a temperature softmax over the cluster scores, at temperature $\tau = 1.0$; the temperature sets how sharply the budget piles onto the worst clusters. **(iv) Fill each quota** by drawing the foregrounds whose masked cutouts sit closest to the funded cluster’s centroid.

The arms, and what each one holds fixed. All arms share the same base pool and differ only in which 1000 images are appended to it, so any difference between two of them is attributable to that choice and nothing else. **A0** appends nothing. **A2** appends 1000 real photographs, which separates the effect of a larger pool from the effect of generated content. **B** appends 1000 renders drawn

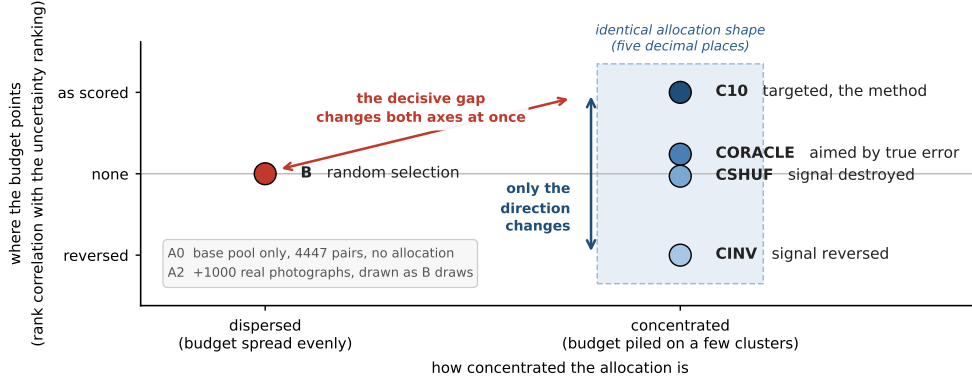


Figure 1: **What each arm holds fixed and what it varies.** The horizontal axis is how concentrated an allocation is; the vertical axis is where it points, measured as rank correlation with the committed uncertainty ranking. Because the C-family are permutations of one allocation, they reproduce its shape to five decimal places and differ only vertically — which is what makes the vertical contrast clean and the diagonal one ambiguous. A0 and A2 append nothing and no generated content respectively, so neither has an allocation to place. *Sources:* `abc_pools.json`; `T2_RESULTS.md`; `results.md` §5.2.

uniformly at random, redrawn per seed. **C10** appends the 1000 that the allocation above selects. A2, B and C10 all hold 5447 pairs, and the comparison this paper turns on is C10 against B.

That comparison cannot settle the question on its own, and Figure 1 is why: a targeted allocation differs from a random one along *two* axes at once — it is more concentrated, and it points somewhere — so a difference between C10 and B is attributable to neither. Three further arms hold concentration exactly fixed and vary only where the budget points. **CSHUF** permutes the cluster scores, destroying the signal while preserving the allocation’s shape; **CINV** reverses their rank; **CORACLE** replaces the score with each cluster’s true endpoint error.

What decides. Write $\hat{\sigma}$ for the seed-to-seed wobble — the pooled within-arm standard deviation of S_α , taken at the conventional $\alpha = 0.5$. A gap is a **REAL EFFECT** only if it exceeds twice that wobble *and* every seed agrees on its sign; a smaller gap is **WITHIN NOISE**; one that clears the bar without agreeing in sign is **INCONCLUSIVE**, and is reported as such rather than resolved by adding seeds. The rule applies per architecture, and a disagreement between them is reported as a disagreement. We report no p -values at three seeds. Appendix B records the rule’s provenance, the seeds, and the gates each campaign passed before launching.

3 RESULTS

Four analyses follow, each measured independently of the others: whether the targeted allocation beats a random one on trained accuracy, what makes a targeted set measurably different regardless of that null, the mechanism behind the null, and a separate finding about the benchmark used to read all of the above.

3.1 DOES TARGETING BEAT RANDOM?

The loop is specified, so the first question is the one it was built to answer. **Aiming the budget by uncertainty did not beat spending it at random.** On the primary endpoint at eight seeds per arm, the gap between the targeted allocation and a random one is $+0.0024$ — about a third of the spread it would have had to clear — and every one of the sixteen gaps that campaign decides returns the same verdict.

A null is only as good as the objections it survives, so the rest of this section is those objections and what each one was answered with. Table 1 is the whole trained record: four campaigns that return

Table 1: **Every trained comparison, on the primary architecture and endpoint.** A gap must exceed its own campaign’s bar *and* agree in sign across seeds to count. Four campaigns attack the decisive gap from different directions and none of them moves it; the positive control, which compares against a method we expected to be clearly worse, clears its bar more than threefold. Per-cell tables for both architectures and both endpoints are in Appendix C. *Sources:* `abc-verdict.json` and its `or/`, `fx/`, `se/` variants; `pc-verdict.json`.

	What it tests	n	Gap	Δ	$2\hat{\sigma}$	$\times$ bar	Verdict
ABC	targeting vs. random, as committed	3	C10–B	+0.0050	0.0179	0.28	WITHIN NOISE
SE	the same gap at eight seeds	8	C10–B	+0.0024	0.0069	0.34	WITHIN NOISE
OR	a perfect score in place of ours	3	CORACLE–B	+0.0008	0.0061	0.14	WITHIN NOISE
FX	the optimisation budget unpinned	3	C10FX–BFX	+0.0014	0.0116	0.12	WITHIN NOISE
PC	<i>can this design see anything?</i>	3	C10–MT	+0.0197	0.0060	3.31	DETECTED

nothing, and one that returns something, which is how we know the fourth column means anything at all.

More seeds made the effect smaller, not the measurement finer. That is the plain form of what the seed expansion returned, and it is worth stating before the arithmetic, because the arithmetic looks backwards. Eight seeds per arm give a truer estimate of the seed-to-seed spread than three did: the real spread is *larger* than the optimistic three-seed guess, while the gap itself halved, from +0.0050 to +0.0024. So the bar sits at 0.0069 on 28 degrees of freedom rather than the 0.0034 the pre-registration projected — wider, on three and a half times the evidence about how noisy this training actually is — and the gap now sits inside even the tightest bar this project has measured, 0.0055, which at three seeds it did not. Appendix C gives the per-arm spreads and the projection this replaces.

Three objections, three measurements. Each of the following is a reason the null might be an artifact rather than a result, and each has a campaign attached rather than a concession.

Perhaps we simply chose a bad uncertainty estimator. The oracle campaign replaces the score with each cluster’s *true* error on the endpoint — information no real method could have — and nothing moves: all eight of its gaps are within noise and not one is even directionally stable. The substitution was not a small one, either. Our score and the truth it stands in for agree at a rank correlation of +0.2372, so we replaced the signal with something several times better aimed and the trained result did not notice. This converts *our score did not help* into the stronger *no score of any quality helped here*.

Perhaps the added images were never really trained on. They were not, under the committed schedule: steps per epoch are set by the shorter of two data loaders, so appended images displace base-pool exposure instead of adding to it. The FX campaign removes that pin, which raises steps per epoch from 253 to 341 on one architecture and 127 to 171 on the other. The decisive gap stays within noise in all four cells and *shrinks*, to +0.0014. Growing the budget did not reveal an effect the pin was hiding.

Perhaps three seeds is too few. It was, and that is what the seed expansion above measures. It is a separate campaign with its own arms, noise estimate and verdicts rather than extra seeds bolted onto a finished one, because adding seeds after seeing a result is how a null gets talked into existence.

The instrument is not blind. A null is only worth reading if the same measurement can detect something real, so we trained an arm we expected to be clearly worse — a mean-teacher baseline, same architecture, same seeds, same scorer, same rule — and asked whether this design would see the difference. It did, comfortably and with every seed agreeing. **And that control does not close the gap that matters most.** It shows an effect of roughly the reference magnitude is visible; it does not show that an effect the size of our nulls would be. The control’s own bar, 0.0060, is itself larger than the decisive gap it is meant to reassure us about, so a positive control at 0.02 licenses *the instrument works, not it would have seen the effect we are nulling*.

What the measurement can and cannot rule out. The decisive statement is the pre-registered one above: no effect on the primary cell larger than the bar the seed expansion measured. A post-hoc interval on the two arms actually compared is tighter still, placing the gap in $[-0.0020, +0.0067]$, which **excludes the reference effect by more than a factor of two** while admitting effects a few times smaller. That interval decides nothing — the frozen rule admits no interval — but it is a stronger honest statement than “we could not have seen it”. At eight seeds a formal equivalence test also becomes informative for the first time, and Section 3.2 reports it where the asymmetry it reveals belongs.

3.2 CONCENTRATION, NOT TARGETING

The null says the direction of the signal did not move the endpoint. It does not say that targeting changes nothing — and it plainly does change something, because a targeted training set is easy to tell apart from a random one. **What makes it different is that the budget is concentrated on a few clusters, not that it is aimed at the right ones.** Three independent measurements say so: the geometry of the selected set, trained accuracy at eight seeds, and the one family of metrics on which the effect of concentration is large enough to resolve.

The geometry: every rule that concentrates reproduces the separation. Aim a generation budget at the clusters where the model is least confident and the training set you get really does look different from a randomly chosen one of the same size — train a probe to tell the two apart and it succeeds easily, on images it has never seen. But re-run the selection with the uncertainty scores shuffled between clusters, so the budget is spread exactly as unevenly while pointing at the wrong places, and the separation survives. Aim at an arbitrary cluster instead: it survives again. Only removing the unevenness itself, by drawing two random sets of equal size, collapses it (Figure 2a). The probe was detecting concentration all along.

Paired across twenty cells of budget and configuration, the real signal contributes $+0.0073$ of a Cohen’s d over its own shuffle — and against an arbitrary cluster it is *negative*, at -0.0649 . On the strongest directional evidence we have, aiming does slightly worse than not aiming. What targeting buys is narrowness rather than reach: an effective-rank ratio of 0.634 in every cell tested, covering no more of the target manifold than a random draw.

Trained accuracy: reversing the signal does not hurt. Geometry is not performance, so we trained it. The falsification arms hold the allocation’s shape fixed to five decimal places and vary only where the budget points, and at eight seeds per arm the arm that *reverses* the uncertainty signal — aiming the budget at the clusters the score calls easiest — has the highest mean score in **four of the four** architecture–endpoint cells (Figure 2b). This is not a claim that reversal helps — every one of those differences is within noise. It is the absence of an ordering: if direction were what mattered, reversing it would cost something, and nothing is paid. Even the destroyed-against-reversed contrast, which carries the strongest sign agreement in the campaign at 8/8 on one cell, does not reach two-thirds of its bar.

At eight seeds a formal equivalence test becomes informative for the first time, and which gaps it resolves is telling. Five of sixteen are equivalent within ± 0.005 , and they are the *direction* contrasts: targeted against its own shuffle rejects at $p = 0.019$, while the targeting-versus-random gap does not reject at all ($p = 0.101$). Direction is equivalent; concentration is not — the same asymmetry, in statistical form. These tests are post-hoc and decide nothing.

Boundary metrics: where concentration is resolvable, reversal captures all of it. The acquisition signal of Equation 1 is boundary-focused, but S_α is structure-aware rather than boundary-specific, so the campaign never measured the quantity the signal was built to move. Re-scoring every run’s committed predictions with two boundary metrics closes that gap (Table 2), and on SINet the concentration effect finally clears its bar. So does the anti-targeted arm, at $1.34\times$ against targeted selection’s $1.37\times$ — the same effect to within a tenth of a bar — while the two direction contrasts stay far inside it. **This is the strongest form of the result we have**, because it is the one place where the same instrument resolves concentration and fails to resolve direction *in the same cells*: the null on direction is not simply an absence of sensitivity.

Table 2: Arm differences on *boundary* metrics, COD10K-test, same pooled- $\hat{\sigma}$ rule. **Post-hoc: not pre-registered, and no verdict is re-decided.** On SINet the concentration gap C10 – B clears its bar — and so does the *anti*-targeted CINV – B, by as much — while both direction contrasts stay within noise. SINet-v2 resolves none of it. *Source:* `diag_subgroups.json`.

Arch.	Metric	$2\hat{\sigma}$	C10–B	CINV–B	C10–CSHUF	C10–CINV
SINet	Boundary IoU	0.0101	+0.0138 3/3	+0.0135 3/3	+0.0038 2/3	+0.0004 2/3
SINet	Boundary F	0.0151	+0.0190 3/3	+0.0173 3/3	+0.0045 2/3	+0.0017 2/3
SINet	IoU	0.0109	+0.0112 3/3	+0.0127 3/3	+0.0019 2/3	–0.0016 2/3
SINet-v2	Boundary IoU	0.0062	+0.0010 2/3	+0.0025 2/3	–0.0002 2/3	–0.0015 3/3
SINet-v2	Boundary F	0.0065	+0.0026 2/3	+0.0050 3/3	–0.0002 2/3	–0.0025 3/3
SINet-v2	IoU	0.0100	–0.0003 2/3	+0.0006 2/3	–0.0013 2/3	–0.0010 2/3

Bold marks a gap exceeding $2\hat{\sigma}$ with 3/3 sign consistency; counts after each Δ are sign-consistency across the three seeds. The pattern is not an artifact of where predictions are binarised — Appendix C gives the threshold sweep.

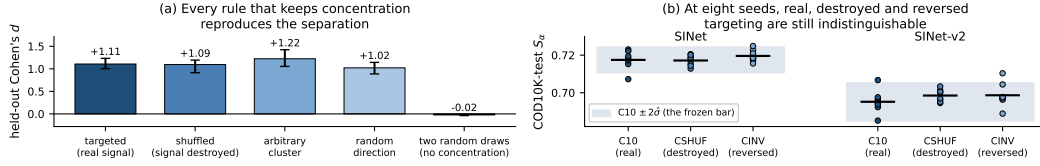


Figure 2: (a) How far a selected set sits from a random set of the same size, in held-out Cohen’s d . Every rule that keeps the allocation’s *concentration* reproduces the separation; only destroying concentration collapses it. Bars are means over four cells, whiskers the range. (b) Trained S_α at eight seeds for the three arms that hold concentration fixed and vary only direction; points are seeds, rules are arm means, the band is the $\pm 2\hat{\sigma}$ bar. *Sources:* `cl_attribution.csv`; `out/se/abc_metrics.csv`.

Three qualifications, all cutting against the result. These metrics were not pre-registered and re-decide no verdict; Table 1 stands. They binarise predictions where S_α does not, though the pattern survives every binarisation threshold we swept. And **SINet-v2 resolves none of it** — which we report rather than leading with the architecture that cooperates.

3.3 WHY IT FAILS

Concentration explains why targeting *looks* like it works. It does not explain why this loop could never have delivered the thing it promised, and that has a mechanism — one chain, measured end to end before any of the training runs above were scored.

The target barely clusters. Under every embedding we tried, k -means separation peaks at a silhouette of 0.1600 and is lower in the other two spaces — far below any conventional threshold for calling a partition real. So a cluster-wise policy is drawing lines the data does not support; **and even where those lines are real, the error does not sit between them.** Only about a fifth of the endpoint’s structural-error variance is between-cluster at all, so four fifths of what a targeted policy would have to fix is invisible to *any* cluster-level allocation, however well aimed.

What the budget does reach is not the endpoint but one half of a mixture. CAMO images are 24.8% of the target pool, and the allocation is not neutral between the two components: 51.5% of the budget lands on them, while the two largest quotas together hold 7 of the 2026 images the paper is scored on. “Allocate by uncertainty” turns out to be, in substantial part, “allocate to CAMO” — **and the quantity doing the allocating is itself the wrong one.** The uncertainty score tracks per-pixel error about twice as strongly as it tracks the structural error the endpoint measures (+0.8553 against +0.4276), and that ordering holds for three independent uncertainty estimators on both architectures, so it is a property of uncertainty-guided allocation here rather than a quirk of our

particular score. Partialling out object area, the most plausible mechanical explanation, *raises* the structural correlation in every row — the opposite of what the confound predicts.

And beneath all of it, the generator supplies nothing new to allocate over. It paints backgrounds and copies objects: the render set is a bijection onto the base foregrounds and not one of 8885 traced objects shows any sign of having been regenerated. The distribution it produces is narrower than its own input — synthetic images reach 0.13 to 0.54 of the real target distribution, while the generator’s own source photographs, real pictures of an entirely different genre, reach 0.71 to 0.75. A policy allocating over that pool is choosing among copies.

One pattern points the other way, and we report it. Pool *size* may matter where pool *content* does not: the unpadded arm trails the padded ones consistently, reaching a REAL EFFECT on the secondary endpoint and two INCONCLUSIVE verdicts once the schedule is unpinned. No single step clears its bar and we claim nothing from it, but it points away from our conclusion, which is why it is here. Separately, we withdrew one of our own candidate causes: we suspected the generator’s conditioning channel was too narrow to act on a detailed instruction, read the released source, and found four routes carrying foreground information with the dominant one bypassing the bottleneck we had blamed — so the explanation was wrong and is retired rather than left standing. Appendix D gives the full clustering sweep, the signal-generality table and the area control.

3.4 CHAMELEON IS MOSTLY TRAINING DATA

The causes above are about our loop. This one is about the benchmark everyone uses, and it holds whether or not our loop worked.

Fifty of CHAMELEON’s 76 images — 65.8% — are the same photographs as images in the COD10K-train and CAMO training pools, re-encoded, rescaled or cropped. Forty-four have their partner in the public COD10K-train split. A model trained on that split has therefore already seen most of CHAMELEON before being evaluated on it. The standard check does not notice: hashing the files returns *zero* overlap, and it is not wrong — the copies were re-saved in a different format, so not one byte matches while every pixel does. Sixteen images resolve neither way and are reported UNCHECKED rather than clean, so 50 is a floor rather than an estimate.

The count is measured in two tiers (Figure 3). The floor is what a released same-dimension detector reproduces on any machine: 41 images whose partner shares their exact pixel dimensions and differs only by re-encoding. A rescaled or cropped copy leaves its dimension group and is invisible to that detector, so we recover those by matching keypoints under a geometric model and then — because agreeing on a homography is not the same as being one photograph — warping each candidate into the CHAMELEON frame and measuring the residual. **Nine survive that test;** one cleared the keypoint threshold, failed the residual, and is excluded. Thresholds are calibrated on pairs whose answer is already committed, the finding is verified against a byte-identical author-sourced release, and the same detector pointed at NC4K flags 0 of 120 (Appendix E).

We claim non-independence, not inflation. We expected the contamination to flatter CHAMELEON scores, measured it, and did not find it: all eight difficulty percentiles for the leaked subset fall between 0.448 and 0.559. A protocol violation does not need a score advantage to disqualify a benchmark. We also withdraw our own earlier recommendation to report CHAMELEON on its uncontaminated subset, because that subset is now 10 images, one of which is byte-identical to an image in the split we select checkpoints on. **A ten-image endpoint is not an endpoint: report COD10K-test and NC4K instead, and disclose the unchecked share.**¹ The detector, the pair list and the clean protocol are released and run without any of our data.

4 DISCUSSION

Two kinds of evidence sit underneath this paper and should not be run together. The *geometric* result is positive, and we state it without hedging: concentration, not the direction of the uncertainty signal,

¹This is not hypothetical for the protocol we build on, which designates the real unlabelled COD10K training set as its target domain. Testing on CHAMELEON, it removes those images from the *source* domain — closing that path but not the target pool, which holds the same 44. We attribute no intent: the source-side removal shows the authors were alert to leakage, and the target-side path is invisible without a re-encode-tolerant check.

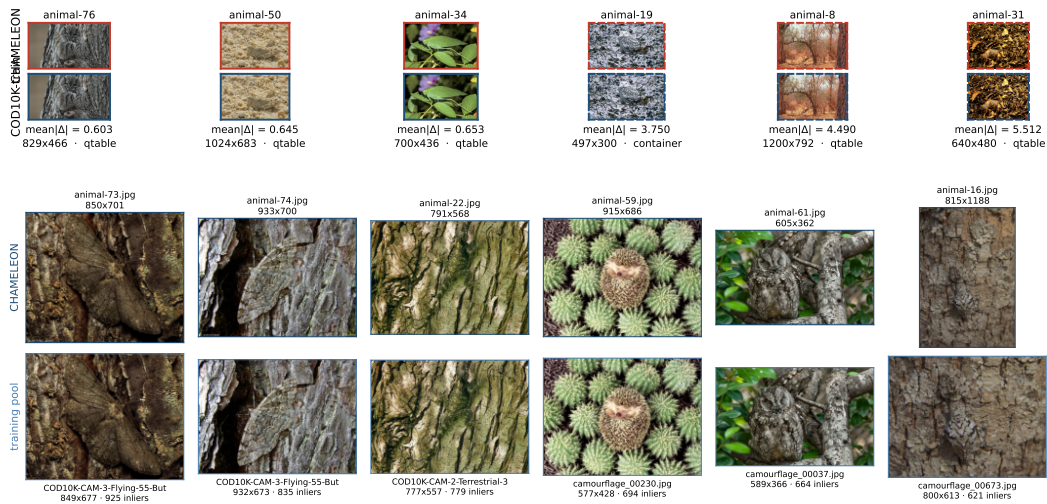


Figure 3: **Two tiers of the same finding.** **Top:** six of the 41 same-dimension re-encodes — the CHAMELEON image above, its training-pool partner below. **Bottom:** six of the 9 recovered by geometric adjudication, which the same-dimension detector reports UNCHECKED because a rescale or a crop moves an image out of its dimension group; note the differing pixel dimensions in each pair’s labels. *Sources:* chameleon_contaminated.json; D2_FINALAUDIT/out/adjudication.csv.

produces the separation that makes targeted selection look like it works. The *downstream* result is a null under stated sensitivity — it does not show that uncertainty guidance cannot work, and we have named the size at which it would have.

Four things transfer beyond this setup. **Three checks cost no training and should precede any cluster-wise allocation:** whether the data separates into clusters at all, what the funded clusters are made of relative to the pool they are drawn from, and how much of the endpoint’s error sits between clusters rather than within them. **The discriminating control is not a random baseline** but a same-shape allocation with the signal destroyed inside it, because random selection differs from targeting in two ways at once. **A null on the endpoint metric does not license a null on the quantity the acquisition signal was built to move.** And a real-versus-synthetic AUC near 1.0 is near-vacuous evidence — a lossy re-save of the identical images separates at 0.9928.

The idea still seems reasonable. For it to have a chance you would need an optimisation budget that grows with the pool rather than one under which added images displace real ones, a foreground supply not already exhausted, a generator wider than its own input, and a target pool that is not a mixture whose components the signal can tell apart. Nothing here says closed-loop generation cannot work, only that this build could not have shown that it does. One pattern runs through the method and the benchmark alike: **a control that returns zero is informative only if it could have returned something else.**

5 CONCLUSION

The question this paper set out to answer was narrow: under a fixed synthetic-data budget, does aiming generation at what the model finds hardest beat spending the same budget at random? Across 103 trained runs and five controls that each close a different way the null could be an artifact, it does not, at the tightest bar this project could measure. The reason is worth carrying forward on its own: a targeted allocation *is* measurably different from a random one, but only because it concentrates the budget, not because it aims it, and that distinction is invisible to any comparison that omits a same-shape control. A second and independently motivated audit found that the field’s oldest camouflage benchmark is two-thirds training data under an exact-byte check that a routine re-encoding defeats.

Together the two results argue for the same practice: before crediting an acquisition rule or trusting a duplicate check, build the version of the test that could have failed, and see whether it does.

AI USE STATEMENT

Generative AI tools were used in the preparation of this work as coding and writing assistants: for drafting and refactoring analysis and experiment-driver code, for drafting and editing prose, and for literature search. They were not used to generate, select, alter or interpret any experimental result, and no number in this paper originates from a language model.

The design decisions that determine every reported verdict — the arms, the endpoints, the seeds, the noise estimate and the four verdict bands — are recorded in pre-registration documents committed before the runs they govern (Section 2), and amendments to them are append-only and dated. Verification does not rest on trusting the tools or the authors: **every numeric value in this paper is traceable to a log block emitted by a committed script, and no result was transcribed by hand.** Appendix D records that mapping, and the released artifacts allow any reported number to be regenerated from primary data. AI-assisted code was reviewed by the authors and is itself committed, so it can be audited rather than taken on trust; where an AI-assisted analysis produced a result that the pre-registered rule did not sanction, it is reported as post-hoc and marked as deciding nothing (Tables 5 and 2).

The authors take full responsibility for the content of this work, including any part drafted with AI assistance.

ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. It uses four public research benchmarks for camouflaged object detection — COD10K Fan et al. (2022), CAMO Le et al. (2019), NC4K Lv et al. (2021) and CHAMELEON Skurowski et al. (2018) — together with the HKU-IS salient-object set as a source pool and the released LAKE-RED Zhao et al. (2024) generator checkpoint. Each is used for non-commercial academic research, which is the use its distribution terms provide for, and each is obtained from its authors’ own release rather than redistributed by us; we release no benchmark imagery, only the derived pair list and detector described below. CHAMELEON in particular is an unpublished collection that has become a de facto standard, which is part of why its provenance is worth auditing at all.

Section 3.4 reports an overlap between two public datasets. We state explicitly what that measurement does and does not support. It is a measurement of pool membership; it attributes no intent to any dataset’s authors, and we draw the direction of provenance as an inference from release chronology rather than as a finding. We measured for a score-inflation figure, did not find one, and do not report one. The one correction we direct at anyone is directed at ourselves: it was our own protocol that adopted CHAMELEON as an evaluation endpoint without checking whether the repository we built on ever sanctioned it. We release the detector so that the measurement can be repeated, contested, or extended by others rather than taken on trust.

REPRODUCIBILITY STATEMENT

The decision rules governing every result are committed before the runs they govern, at the commits named in Section 2, and amendments to them are append-only and dated. Each experiment has a committed result document and one or more committed log blocks; each number in this paper is traceable to such a block or to a committed artifact, and Appendix D gives that mapping. Per-run metrics for all 103 training runs, the full clustering sweep, and the full signal-generalizability table are in the appendix. The contamination detector is released with an eight-assertion known-answer self-test that runs without any of our data, alongside the confirmed pair list, the clean protocol, and the calibration of the geometric extension.

Artifacts. Code, pre-registration documents, per-run metrics, the append-only experiment log, and the released detector accompany this submission as supplementary material, which preserves anonymity without a third-party host. The supplement is self-contained: every experiment is one

directory holding its scripts, its declared thresholds and its outputs, and each cites the log block that produced it.

Two limits on reproducibility. Our regeneration check establishes byte-identical reproduction on one machine with one software stack; it is not a claim of determinism across hardware or driver versions. And a render from this generator is a function of its foreground, its mask, *and its position within the processing shard*, so changing the shard count changes the outputs even at a fixed seed.

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A ADDITIONAL RELATED WORK

COD benchmarks and detectors. The standard COD evaluation suite comprises COD10K Fan et al. (2022), a large benchmark with ten camouflage categories and pixel-accurate masks; CAMO Le et al. (2019), which focuses on man-made camouflage and occlusion; NC4K Lv et al. (2021), assembled from Flickr; and CHAMELEON Skurowski et al. (2018), an earlier unpublished collection that has become a de facto standard. The continued use of all four matters for Section 3.4: an image shared by a training set and any reported test set contaminates evaluation for methods using the full suite. SINet Fan et al. (2020) decomposes COD into search and identification branches, while SINet-v2 Fan et al. (2022) adds a neighbour connection decoder for boundary refinement; we use both off-the-shelf. SegMaR Jia et al. (2022) iteratively magnifies and re-processes ambiguous regions, representing architectural progress outside our allocation experiment. We report the structure-aware S-measure Fan et al. (2017), enhanced-alignment E-measure Fan et al. (2018), and weighted F_{β}^w Margolin et al. (2014) for comparability with S2R-COD.

Synthetic generation and domain adaptation. Latent diffusion Rombach et al. (2022) and DDIM Song et al. (2021) provide the high-resolution generator and deterministic sampling schedule inherited by our pipeline. Copy-paste augmentation Ghiasi et al. (2021) similarly composes object cut-outs with varied backgrounds; LAKE-RED applies this compositional idea while retrieving background patches. CyCADA Hoffman et al. (2018), in contrast, aligns source and target feature distributions with cycle-consistent adversarial training; our loop does not operate at the feature level and instead feeds generator output into a self-training objective. Other camouflage-image generators exist Chen et al. (2025), but they are outside the experiment, so the reported null is bounded to LAKE-RED.

Uncertainty-guided acquisition and self-training. FixMatch Sohn et al. (2020) couples confidence thresholding with consistency regularisation; our pseudo-labels likewise arise in an inter-round consistency step, but drive a second training round rather than an online update. Pseudo-labelling Lee (2013) supplies the hard-target view of this step. DINOv2 Oquab et al. (2024) and

Table 3: Where this work sits among systems that steer a synthetic-data budget. *Acquisition signal* is what decides where the budget goes; *generator control* is what the method can actually vary in the generated image; *diversity control* is what stops the budget piling onto near-duplicates; *closed-loop* means the downstream model’s state feeds back into generation; *abstention / control* is whether a signal-destroying or same-shape control is run, which is what separates an effect of aiming from an effect of concentrating. The last column is the one this paper argues is missing from the literature, not a claim of superiority — our own entry records a null.

Work	Task	Acquisition signal	Generator control	Closed-loop	Abstention / control
LAKE-RED Zhao et al. (2024)	COD synthesis	none (uniform)	background only	no	—
S2R-COD Luo et al. (2025)	COD adaptation	none (uniform)	background only	pseudo-labels	—
GAUDA Frisch et al. (2025)	surgical seg.	epistemic, per class	image + mask	yes, on-line	none reported
DisCL Liang et al. (2025)	long-tail cls.	hardness, per sample	guidance strength	curriculum	none reported
SynQuE Chen & Zhong (2025)	dataset ranking	proxy score, per set	n/a (picks sets)	no	none reported
This work	COD adaptation	disagreement, per cluster	background only	pseudo-labels	same-shape shuffle, rank reversal, oracle

CLIP Radford et al. (2021) provide the frozen representations used for clustering. These choices complement core-set selection Sener & Savarese (2018): the clusters spread the budget, while the disagreement scores determine which clusters receive more of it. BADGE Ash et al. (2020) jointly uses uncertainty and diversity in gradient-embedding space; our same-shape shuffle in Section 3.2 is a direct control for the corresponding entanglement in this allocation.

Benchmark integrity and negative results. Dataset-collection procedures can induce shared statistical bias Torralba & Efros (2011), and near-duplicate analyses of CIFAR and ImageNet show that visually identical images can cross split boundaries Barz & Denzler (2020). Related contamination concerns appear in language-model evaluation Wu et al. (2024); the mechanism differs here, but the need for explicit auditing is shared. Our pre-registration follows reproducibility guidance Pineau et al. (2021) and the case for publishing disconfirming results Karl et al. (2024). Finally, criticisms of learned comparison measures under resizing and representation choices Parmar et al. (2022); Jayasumana et al. (2024) motivate treating encoding-sensitive probes as diagnostics rather than as clean perceptual evidence.

Where this work sits among budget-steering systems. Table 3 places the three systems discussed in Section 1.1 beside this one. The last column is the point: it records whether a signal-destroying or same-shape control is run, which is what separates an effect of *aiming* a budget from an effect of *concentrating* it. It is not a claim of superiority — our own entry records a null.

B DETAILED METHOD SPECIFICATION

Training. SInet (Fan et al., 2020) at batch 16 for 40 epochs and SInet-v2 (Fan et al., 2022) at batch 32 for 100 epochs, Adam at 10^{-4} , images at 352×352 with ImageNet normalisation. The EMA teacher uses momentum 0.996 under this task setting. The consistency weights in Equation 1 are the configured $a = 0.9$, $b = 0.3$ with the unweighted cross-entropy variant — these are the values the trainer sets for this task, not the loss class’s own defaults. Strong views add random autocontrast and Gaussian blur to the weak view. `cudnn.deterministic` is enabled and `cudnn.benchmark` disabled, which reduces, and is not claimed to eliminate, kernel nondeterminism.

Seeds and what was held constant. The committed seeds are $\{42, 43, 45\}$; the seed expansion of Section 3.1 adds $\{46, 47, 48, 49, 50\}$, fixed in writing before its first run. Across the four-arm campaign’s 24 runs the base pool is asserted byte-identical against a hash manifest with 0 mismatches, both rounds read the unfiltered 4040-image target set, and every run shares one trainer invocation, checkpoint rule, endpoint set and deterministic-kernel setting.

Optimisation budget. Steps per epoch are set by the shorter of the source and target loaders, which pins them at 253 (SINet) and 127 (SINet-v2) in both rounds of every committed run regardless of pool size, so appended images displace base-pool exposure rather than adding to it. This is measured per run and asserted rather than assumed. It is not an asymmetry between the arms that matter — B and C10 hold identical pool sizes and therefore receive identical step counts — and Section 3.1 reports the campaign that removes the pin and re-runs the comparison without it.

Constructing the falsification arms. CSHUF’s permutation is the **first** of 64 pre-declared draws satisfying zero fixed points and $|\rho| \leq 0.10$, chosen by a rule fixed before the draw and never by inspecting its consequences; its realised rank correlation with the true score vector is -0.03351 , against CINV’s exact -1.0 . CINV’s single fixed point is mathematically necessary: at odd k the median-ranked cluster is its own mirror. Because equality is the *supporting* outcome for these arms, two near-identical arms would manufacture it, so set distinctness was gated before any GPU time at a Jaccard bound of 0.50: between 452 and 511 of each pair’s 1000 injected images differ, at $2.2\text{--}2.4\times$ chance overlap.

Allocation sweeps. Stage (iv) matches against a grey-masked cutout rather than a render, because at selection time the foreground exists and the render does not. The budget grid is $B \in \{250, 500, 1000, 2000, 3000, 4447\}$ and the temperature grid over τ is extended until both degenerate ends are reached — concentrated when $\max_j p_j \geq 0.95$, uniform when total variation from uniform falls below 0.01 — with caps at $\tau \in [10^{-4}, 10^3]$. The campaign uses $\tau = 1.0$, $B = 1000$.

Metrics. S_α at $\alpha = 0.5$ is the primary endpoint metric; MAE is the checkpoint-selection metric; F_β^w and E_ϕ are computed and reported but decide nothing. The F -measures apply β in the non-squared form, so the metric name rather than a formula is what identifies them.

Degrees of freedom. Each campaign’s $\hat{\sigma}$ is pooled within-arm across that campaign’s own arms, at $\text{df} = (\text{arms}) \times (\text{seeds} - 1)$: 8 for the three-seed campaigns over four arms, 6 for the three-arm controls, and 28 for the eight-seed expansion. A campaign’s bar is never swapped for another campaign’s in either direction.

The A2 control’s one confound. A2 pairs the authors’ original photographs with the authors’ masks, whose mean foreground fraction differs from the raw masks by 0.0058 on 1000 of 5447 images. This affects $\Delta(B - A2)$ and is disclosed; it does not touch $\Delta(C10 - B)$, the decisive gap, since B and C10 share mask provenance, pool size and render pool.

Pre-registration commits and amendments. Every decision rule was committed before the runs it governs. The four-arm rule is recorded at 065dac6, the commit carrying that campaign’s first log block; the falsification campaign’s at 617a2e1; the signal-generality check’s at 6efde9f; the area control’s and the positive control’s at c2114af; the seed expansion’s at b755aa9, dated 2026-09-18. The oracle and optimisation-schedule campaigns carry their own pre-registrations, each committed before that campaign’s first training run. Three amendments exist, all append-only and dated. Two are reference-tolerance defects of the same kind: a frozen clause requiring reproduction to better than 10^{-9} against a file recording six decimal places, and — in the area control, written by the same author who had just documented the first — the identical mistake against a file recording four. Both were replaced by exact agreement at the recorded precision, which is the strictest satisfiable form, and neither altered a decision threshold. The third marks one verification gate not applicable in scope after it was found to compare two architectures’ signals.

C THE DECISIVE GAP IN EVERY CELL, AND THE BARS IT IS READ AGAINST

Table 1 reports the primary cell. Table 4 reports all four, for each campaign that attacks the decisive gap. **Sixteen of sixteen are WITHIN NOISE, and not one reaches half of its bar.**

Table 4: The decisive gap in all four architecture–endpoint cells, per campaign. Each campaign is read against its *own* pre-registered bar, never another’s. *Source:* `abc_verdict.json` and its `se/`, `or/`, `fx/` variants.

	Cell	n	Gap	Δ	$2\hat{\sigma}$	$\times$ bar	sign
ABC	SINet · COD10K	3	C10–B	+0.0050	0.0179	0.28	3/3
	SINet · NC4K			+0.0021	0.0083	0.25	3/3
	SINet-v2 · COD10K			−0.0004	0.0123	0.03	2/3
	SINet-v2 · NC4K			+0.0009	0.0105	0.09	2/3
SE	SINet · COD10K	8	C10–B	+0.0024	0.0069	0.34	6/8
	SINet · NC4K			+0.0003	0.0059	0.06	4/8
	SINet-v2 · COD10K			−0.0011	0.0101	0.11	5/8
	SINet-v2 · NC4K			−0.0009	0.0093	0.10	4/8
OR	SINet · COD10K	3	CORACLE–B	+0.0008	0.0061	0.14	2/3
	SINet · NC4K			−0.0012	0.0047	0.26	2/3
	SINet-v2 · COD10K			+0.0023	0.0093	0.25	2/3
	SINet-v2 · NC4K			+0.0020	0.0080	0.25	2/3
FX	SINet · COD10K	3	C10FX–BFX	+0.0014	0.0116	0.12	2/3
	SINet · NC4K			+0.0023	0.0107	0.21	2/3
	SINet-v2 · COD10K			−0.0007	0.0100	0.07	1/3
	SINet-v2 · NC4K			−0.0003	0.0122	0.03	1/3

Why the committed bar is so wide, and why we did not narrow it. The four-arm campaign pools its within-arm spread across all four of its arms, and one of them dominates. On the primary cell the per-arm standard deviations are A0 0.01727, A2 0.00193, B 0.00181 and C10 0.00406: **A0 alone contributes 92.7% of the pooled variance**, and A0 appears in the decisive comparison not at all. Its instability is unexplained and is reported rather than excluded, because the bar was pre-registered as a pooled quantity and re-deriving it after seeing the result is precisely the move the pre-registration exists to prevent. What we do instead is report the tighter bars that later campaigns measured on their own arms, and let the reader see all of them.

The three bars, and what each is good for. Three estimates of the same wobble exist, and they are not interchangeable. The committed four-arm bar is 0.0179 and is inflated by A0 as above. The falsification campaign, pooling over B, C10, CSHUF and CINV — the arms that actually appear in a decisive gap — measures 0.0055 at the same degrees of freedom. The seed expansion measures 0.0069 over the same four arms at 28 degrees of freedom instead of 8. The asymmetry that matters is this: where equality is the *supporting* outcome, a tighter bar is conservative, but here the claim and the null coincide, so a looser bar flatters us. That is why the paper reads the gap against the tightest defensible bar rather than the widest, and why the seed expansion’s bar — not the committed one — is the sensitivity statement in Section 3.1.

The bar widened when the seeds grew, and that is the expected behaviour. The seed expansion’s pre-registration projected that eight seeds would tighten the bar by $\sqrt{8/3} \approx 1.63\times$, *if the per-arm spreads stayed as measured*. They did not. Every one grew: on the primary cell B went $0.00181 \rightarrow 0.00251$, C10 $0.00406 \rightarrow 0.00489$, CSHUF $0.00166 \rightarrow 0.00278$ and CINV $0.00285 \rightarrow 0.00302$. A pooled standard deviation on eight degrees of freedom is a noisy estimate that lands low as often as high; on twenty-eight it does not. So the expansion did not buy a smaller bar, it bought a trustworthy one — and in doing so showed the three-seed bars to have been optimistic rather than conservative.

Table 5: Post-hoc interval estimates at **eight seeds per arm**, COD10K-test S_α . *These decide nothing*: verdicts remain the pre-registered 2δ rule of Section 2. Intervals are Welch on the two arms actually compared, so they are not inflated by an arm outside the comparison. TOST is run at $\delta = 0.005$; bold marks the comparisons that reject, and the final column gives the smallest δ at which each comparison *could* have concluded equivalence. *Source*: `out/se/abc_metrics.csv`, via `figs/make_tost_table.py` using `make_results.py`’s `welch_tost`.

Arch.	Gap	Δ	95% CI	TOST p	smallest δ
SINet	C10 – B	+0.0024	$[-0.0020, +0.0067]$	0.1009	0.0059
SINet	C10 – CSHUF	+0.0003	$[-0.0041, +0.0047]$	0.0189	0.0039
SINet	C10 – CINV	-0.0021	$[-0.0066, +0.0023]$	0.0905	0.0057
SINet	CSHUF – CINV	-0.0024	$[-0.0055, +0.0007]$	0.0486	0.0050
SINet-v2	C10 – B	-0.0011	$[-0.0067, +0.0045]$	0.0764	0.0057
SINet-v2	C10 – CSHUF	-0.0033	$[-0.0087, +0.0020]$	0.2516	0.0077
SINet-v2	C10 – CINV	-0.0035	$[-0.0101, +0.0031]$	0.3143	0.0089
SINet-v2	CSHUF – CINV	-0.0001	$[-0.0057, +0.0054]$	0.0402	0.0047

At three seeds no comparison reached equivalence at $\delta = 0.005$; at eight, **the direction contrasts do and the targeting-versus-random gap does not**. That asymmetry is the same one Section 3.2 reports, in statistical form.

The interval at three seeds, and why it left the main text. A Welch interval on the two arms actually compared, at the committed three seeds, places the decisive gap in $[-0.0035, +0.0136]$ and excludes the reference effect. It is superseded by the eight-seed interval reported in Section 3.1 and is recorded here only for continuity with the committed account: at roughly 2.8 degrees of freedom its exclusion held by a margin too thin to lean on, which is exactly the weakness the seed expansion removes.

Boundary metrics: the binarisation sweep in full. Boundary metrics binarise a prediction where S_α does not, so the result in Section 3.2 could in principle be an artifact of where the threshold sits. It is not. Swept over thresholds 0.4/0.5/0.6 on SINet Boundary-IoU, $\Delta(\text{C10} - \text{B})$ holds at $1.39/1.37/1.23\times$ its bar and $\Delta(\text{CINV} - \text{B})$ at $1.25/1.34/1.26\times$, both 3/3 sign-consistent at every threshold, while the two direction contrasts never exceed $0.52\times$ and $0.14\times$ anywhere in the sweep. Concentration resolves at every threshold; direction resolves at none.

How much the trained comparison is attenuated. The trained half of the concentration result is diluted three times over, and we disclose it because the attenuation biases toward the null we report. Injected images are 1000 of 5447 pairs, or 18.4% of the pool; under the committed schedule the pinned step count dilutes their exposure further; and only about 48% of injected images differ between any two arms, giving an effective contrast near 8.8%. Two of the three are now measured rather than argued: the optimisation-schedule campaign removes the pin and the decisive gap *shrinks*, and the seed expansion triples the degrees of freedom behind the bar. What remains is the third, which is a property of comparing two arms that share 89% of their training data by construction.

D SUPPLEMENTARY TABLES

E CHAMELEON CONTAMINATION: CALIBRATION, CONTROLS AND ADJUDICATION

Tier A — the released same-dimension detector. Images are grouped by exact pixel dimensions, since a re-encode preserves them. Within a group each image is reduced to a 32×32 greyscale descriptor, pairs are shortlisted by descriptor RMS distance at 14.0, and survivors are decoded at full resolution and confirmed when the mean absolute difference over RGB falls below 6/255. We deliberately do *not* contrast-normalise the descriptor: normalising destroys the scale that separates a re-encode from a merely similar photograph, and it is what made our own first pass report 10 of 76 rather than 41. Sweeping the tolerance over 1, 2, 3, 5, 6 gives 11, 26, 37, 40 and 41 — the count

Table 6: Full clustering sweep over the target set: silhouette (mean over 10 k -means seeds), bootstrap ARI and seed ARI, for every k in the declared grid and all three embedding spaces. The principled k per space is **bold**. No k in any space reaches a silhouette conventionally regarded as well-separated structure, and clipL224 falls monotonically, so its selection criterion returns the grid edge rather than an interior maximum — recorded as a FAIL of the declared guard. *Source:* `b1_k_sweep_*.csv`; block EXP B1 #3 @ `results/REBUILD_LOG.txt`:922.

	dinoL224			dinoL518			clipL224		
k	sil.	bs.ARI	sd.ARI	sil.	bs.ARI	sd.ARI	sil.	bs.ARI	sd.ARI
5	0.0538	0.712	0.795	0.0566	0.624	0.732	0.0568	0.953	0.983
10	0.0868	0.676	0.709	0.0923	0.598	0.770	0.0550	0.692	0.805
15	0.1053	0.608	0.714	0.1152	0.573	0.592	0.0486	0.575	0.670
20	0.1180	0.629	0.693	0.1326	0.506	0.524	0.0486	0.635	0.646
30	0.1378	0.596	0.642	0.1527	0.539	0.617	0.0488	0.609	0.631
50	0.1465	0.610	0.615	0.1556	0.579	0.600	0.0448	0.498	0.549
75	0.1457	0.588	0.627	0.1600	0.579	0.616	0.0396	0.463	0.477
100	0.1419	0.608	0.637	0.1557	0.583	0.601	0.0382	0.419	0.452
150	0.1367	0.630	0.636	0.1507	0.618	0.625	0.0357	0.380	0.409

Table 7: Full signal-generality measurement, all 16 rows: four signal variants across three signal families, two architectures, two aggregations. ρ at the committed seed, $\pm$ sd over 10 k -means seeds. The whole-image block passes the pre-registered ordering 8/8; the boundary block passes 2/8 with four rows going negative, an outcome outside the pre-registered interpretation space, reported as it fell. *Source:* `rebuild/T2C/out/t2c-table.csv`; block EXP T2C @ `results/REBUILD_LOG.txt`:2550.

arch	signal	agg.	$\rho(\text{MAE})$	$\rho(1 - S_\alpha)$	$\rho(1 - \text{IoU})$	order	seeds
SINet-v2	ES	whole	$+0.7070 \pm 0.0404$	$+0.5628 \pm 0.0359$	$+0.4591 \pm 0.0315$	PASS	10/10
SINet-v2	ensemble A0	whole	$+0.6191 \pm 0.0675$	$+0.5483 \pm 0.0409$	$+0.4540 \pm 0.0451$	PASS	9/10
SINet-v2	ensemble CSHUF	whole	$+0.5576 \pm 0.0690$	$+0.4488 \pm 0.0580$	$+0.3370 \pm 0.0577$	PASS	10/10
SINet-v2	entropy	whole	$+0.7393 \pm 0.0309$	$+0.5404 \pm 0.0408$	$+0.4070 \pm 0.0395$	PASS	10/10
SINet-v2	ES	bound	-0.5485 ± 0.0745	-0.1146 ± 0.0675	$+0.0537 \pm 0.0558$	FAIL	0/10
SINet-v2	ensemble A0	bound	-0.1960 ± 0.0524	$+0.0425 \pm 0.0410$	$+0.0933 \pm 0.0396$	FAIL	0/10
SINet-v2	ensemble CSHUF	bound	-0.2527 ± 0.0668	-0.0591 ± 0.0609	-0.0011 ± 0.0579	FAIL	0/10
SINet-v2	entropy	bound	$+0.4208 \pm 0.0356$	$+0.4182 \pm 0.0455$	$+0.3605 \pm 0.0470$	PASS	8/10
SINet	ES	whole	$+0.6284 \pm 0.0381$	$+0.3433 \pm 0.0301$	$+0.2613 \pm 0.0383$	PASS	10/10
SINet	ensemble A0	whole	$+0.6028 \pm 0.0466$	$+0.2698 \pm 0.0487$	$+0.1491 \pm 0.0560$	PASS	10/10
SINet	ensemble CSHUF	whole	$+0.5212 \pm 0.0564$	$+0.4006 \pm 0.0459$	$+0.3361 \pm 0.0447$	PASS	10/10
SINet	entropy	whole	$+0.5892 \pm 0.0368$	$+0.4267 \pm 0.0277$	$+0.3588 \pm 0.0351$	PASS	10/10
SINet	ES	bound	-0.0012 ± 0.0481	-0.0567 ± 0.0500	-0.0554 ± 0.0479	FAIL	8/10
SINet	ensemble A0	bound	$+0.3633 \pm 0.0605$	$+0.2910 \pm 0.0509$	$+0.2215 \pm 0.0513$	PASS	10/10
SINet	ensemble CSHUF	bound	$+0.3175 \pm 0.0637$	$+0.3857 \pm 0.0381$	$+0.3770 \pm 0.0322$	FAIL	2/10
SINet	entropy	bound	$+0.2434 \pm 0.0414$	$+0.2492 \pm 0.0540$	$+0.2432 \pm 0.0563$	FAIL	6/10

saturates. The detector passes an eight-assertion synthetic known-answer test that uses no repository data.

Tier B — calibration of the geometric extension. SIFT correspondences, Lowe ratio 0.75, filtered by a RANSAC homography at 4 px. Thresholds are fixed on data whose answer is already committed, before the unchecked images are scored: positives are the 41 Tier-A pairs, negatives 4000 random CHAMELEON $\times$ training pairs.

Table 8: The area control (EXP AC). First-order partial Spearman on the same cluster partition and floors as the committed measurement. z_{obj} = per-cluster mean object area on the endpoint side; z_{unc} = per-cluster mean uncertain-region area on the target side. Ordering survives 7/8 and 6/8 against a floor of 6/8 declared before the run. Note that partialling object area *raises* $\rho(1 - S_\alpha)$ in every row — the opposite of the confound’s prediction. *Source:* rebuild/AC/out/ac.table.csv; block EXP AC @ results/REBUILD-LOG.txt:2612.

arch	signal	raw		$ z_{\text{obj}} $		$ z_{\text{unc}} $	
		MAE	$1 - S_\alpha$	MAE	$1 - S_\alpha$	MAE	$1 - S_\alpha$
SINet	ES	+0.6284	+0.3433	+0.5761	+0.4988	+0.5518	+0.3897
SINet	entropy	+0.5892	+0.4267	+0.5505	+0.5517	+0.5060	+0.4426
SINet	ensemble A0	+0.6028	+0.2698	+0.5411	+0.4301	+0.5603	+0.2583
SINet	ensemble CSHUF	+0.5212	+0.4006	+0.5002	+0.4857	+0.4857	+0.3937
SINet-v2	ES	+0.7070	+0.5628	+0.7062	+0.6272	+0.4457	+0.4744
SINet-v2	entropy	+0.7393	+0.5404	+0.7134	+0.6546	+0.5369	+0.4266
SINet-v2	ensemble A0	+0.6191	+0.5483	+0.6245	+0.5969	+0.4355	+0.4488
SINet-v2	ensemble CSHUF	+0.5576	+0.4488	+0.5344	+0.5209	+0.4482	+0.3615
ordering holds		8/8		7/8		6/8	

Inliers on the 41 confirmed pairs	525–1174 (median 1129)
Inliers on 4000 random pairs	max 8
Inliers anywhere in the NC4K control	max 19
Operating point	20
Recall on the knowns at that point	41/41
NC4K flagged at that point	0 of 120

The operating point is one inlier above the largest count *any* negative set produced, which is the NC4K maximum of 19 rather than the random-pair maximum of 8. We also ran resolution-normalised global matching as a second instrument; no unchecked image passed it (lowest normalised RMS 16.7 against a calibrated threshold of 1.02), so every Tier-B match rests on the key-point stage alone.

Adjudication — from “the keypoints agree” to “the same photograph”. An inlier count shows that local patches agree on a geometric model. Two exposures of one static scene agree on a homography as well, so the inlier count alone cannot carry the claim that these files are the same photograph. For each candidate we recover the homography, warp the partner into the CHAMELEON frame and measure the residual over the valid overlap, and we decompose the homography into scale, rotation, shear and perspective. Run on 8 of the 41 known pairs the instrument returns their recorded mean absolute difference to three decimals ($0.603 \rightarrow 0.603$, $5.512 \rightarrow 5.514$) at $H = I$, which is what licenses reading its output on the rest.

CHAMELEON	Inliers	Residual	Corr.	Scale	Partner pool
animal-16	2015	3.76	0.99	1.02	CAMO
animal-73	3332	3.93	1.00	1.00 (crop)	COD10K-train
animal-74	3403	4.17	1.00	1.00 (crop)	COD10K-train
animal-22	3097	5.84	1.00	1.00 (crop)	COD10K-train
animal-72	396	6.60	0.99	6.36	CAMO
animal-61	1456	8.92	0.97	0.98	CAMO
animal-43	263	9.54	0.96	0.53	CAMO
animal-66	860	10.72	0.97	0.51/0.39	COD10K-train
animal-59	1017	12.11	0.97	1.52	CAMO
animal-7	25	rejected — shear -86.5° , scale 0.24 vs 1.24; no homography at full resolution			

Residual is mean $|\cdot|$ over the overlap on a 0–255 scale. Shear and perspective are indistinguishable from zero for all nine. animal-7 was the only candidate below the calibration band of 525–1174

inliers, clearing the operating point by 5 while the NC4K negatives reached 19; its homography collapses the frame towards a line, and it is excluded.

Scope of the exact-hash zero. The 0 we report is measured against the training pool. Across benchmarks it is not zero: `animal-17`, `animal-25` and `animal-60` are byte-identical to CAMO-250 images. Those files are absent from the training pool and so do not enter the contamination count, but CAMO-250 is the split this repository selects checkpoints on.

What remains uncounted. Sixteen images have no partner at either tier and are reported UNCHECKED, not clean. Different photographs of one specimen, montages and heavy colour edits are not detected and are not claimed. Every figure here is a lower bound.

F CHAMELEON MASK RELEASES AND DETECTOR VALIDATION

The two circulating CHAMELEON ground-truth releases are different annotations, unrelated to contamination or to any method. They disagree in polarity, giving a mean IoU of 0.0 as stored and 0.6932 after polarity alignment; only 27 of 76 masks are exactly identical after alignment and 40 of 76 reach $\text{IoU} \geq 0.9$. On identical predictions the choice of release alone moves MAE by $2.7 \times$ (0.2196 against the author-sourced masks, 0.0816 against the repackaged ground truth). Any CHAMELEON column is therefore not comparable across papers unless the release is named.

Detector self-test. The released detector passes an eight-assertion synthetic known-answer test using no repository data, reproduces the canonical 41 pairs and 41 names exactly, and surfaced one additional pair — the second partner of the one image that has two.

G THE THREE CONTROLS: EXECUTION RECORDS AND DISCLOSURES

Section 3.1 closes three objections with three campaigns. Each carries a disclosure that belongs with it.

The oracle campaign uses test labels, deliberately. CORACLE takes the committed allocation’s shape and re-assigns it across clusters by the rank of each cluster’s *true* error on the endpoint. That is information no real method could have, and the pre-registration says so before the first run: it is a diagnostic upper bound and never a method, and a pre-flight gate that would forbid this for any real arm is passed on a technicality, which we disclose rather than claim as cleanliness. One further limitation: the oracle is derived from one architecture’s errors and applied to both. The two architectures’ per-cluster errors agree at a rank correlation of 0.9211, so the second architecture receives a near-oracle rather than a perfect one.

The schedule campaign bundles four changes, not one. Unpinning the optimisation budget does not isolate a single variable. It raises steps per epoch from 253 to 341 (SINet) and 127 to 171 (SINet-v2); it re-uses the same 4040 target images more often, so each is seen about 1.35 times per epoch instead of once; it leaves the learning-rate schedule unrescaled, because rescaling would re-impose a fixed budget and defeat the control; and it therefore changes total optimisation as well as its distribution. Growing the budget by roughly a third made every arm slightly *worse* — 11 of 12 cross-schedule comparisons are negative — most plausibly because the learning rate decays per epoch and was deliberately left alone. A gate declared in advance that any run whose step count stayed at the old pinned value was a halt rather than a result.

The seed expansion is a separate campaign, and survived a power loss. It is not the committed campaign with more seeds: adding seeds after seeing a result is optional stopping, and it would invalidate every verdict the original campaign reports. SE has its own four arms, its own noise estimate, its own bar and its own verdicts, and its eight seeds were fixed in writing before its first run. The twelve runs at the committed seeds are re-scored rather than re-trained, and must reproduce their committed values to six decimal places or the campaign halts.

A host power loss on 2026-09-20 interrupted training. Two runs were killed part-way through their first round; the driver’s own pre-launch check discarded both and re-ran them from scratch, and they are the only two entries in the discard record. No partially trained model enters the scoring pool, no seed was added or dropped, and no SE number existed at the moment resumption was decided — the metrics file was absent, so nothing had been computed and nothing seen. The campaign cost 75.9 GPU-hours across its 40 trained runs.

What differs between arms beyond the injected images. The pseudo-labelling step between rounds selects target images using each arm’s *own* first-round model, so each arm appends a different number of differently-labelled images. Across the seed expansion the means are 1977 (B), 2025 (C10), 1990 (CSHUF) and 1979 (CINV), with per-run values spanning 1817 to 2259. This is legitimately part of “closed loop” and is reported rather than absorbed: it is a second place the arms differ beyond the injection, and it is high-variance. The step count stays pinned regardless, which is asserted per run.

Table 9: Per-run metrics at full recorded precision for every training run in this work — the two frozen campaigns and the mean-teacher positive control — on both endpoints. Arms B and C10 appear once; they were re-scored, not re-trained, for the falsification campaign and reproduced their committed values exactly at six decimal places. *Source:* `rebuild/ABC/out/abc_metrics.csv` and `out/t2/abc_metrics.csv`; blocks EXP ABC #3 and EXP T2 #3.

arch	arm	seed	endpoint	S_α	MAE
SINet	A0	42	COD10K	0.715086	0.074960
SINet	A0	43	COD10K	0.681706	0.077044
SINet	A0	45	COD10K	0.706058	0.077347
SINet	A2	42	COD10K	0.705507	0.075994
SINet	A2	43	COD10K	0.708334	0.078489
SINet	A2	45	COD10K	0.709196	0.080858
SINet	B	42	COD10K	0.711554	0.076085
SINet	B	43	COD10K	0.713632	0.076278
SINet	B	45	COD10K	0.715155	0.075917
SINet	C10	42	COD10K	0.717165	0.073184
SINet	C10	43	COD10K	0.723034	0.075136
SINet	C10	45	COD10K	0.715243	0.078355
SINet	CINV	42	COD10K	0.722612	0.073910
SINet	CINV	43	COD10K	0.717343	0.077323
SINet	CINV	45	COD10K	0.718084	0.073607
SINet	CSHUF	42	COD10K	0.718602	0.074840
SINet	CSHUF	43	COD10K	0.716974	0.077678
SINet	CSHUF	45	COD10K	0.715291	0.079340
SINet	MT	42	COD10K	0.699633	0.093377
SINet	MT	43	COD10K	0.699167	0.087571
SINet	MT	45	COD10K	0.697503	0.091322
SINetv2	A0	42	COD10K	0.688813	0.091691
SINetv2	A0	43	COD10K	0.678740	0.095147
SINetv2	A0	45	COD10K	0.699883	0.085756
SINetv2	A2	42	COD10K	0.691914	0.089186
SINetv2	A2	43	COD10K	0.688977	0.094524
SINetv2	A2	45	COD10K	0.695414	0.087455
SINetv2	B	42	COD10K	0.698340	0.092242
SINetv2	B	43	COD10K	0.689481	0.089883
SINetv2	B	45	COD10K	0.697385	0.087132
SINetv2	C10	42	COD10K	0.692321	0.093395
SINetv2	C10	43	COD10K	0.696328	0.090749
SINetv2	C10	45	COD10K	0.695360	0.089182
SINetv2	CINV	42	COD10K	0.697008	0.088438
SINetv2	CINV	43	COD10K	0.696331	0.087293
SINetv2	CINV	45	COD10K	0.697605	0.090189
SINetv2	CSHUF	42	COD10K	0.701269	0.087506
SINetv2	CSHUF	43	COD10K	0.694490	0.087305
SINetv2	CSHUF	45	COD10K	0.696218	0.089485
SINet	A0	42	NC4K	0.765121	0.085752
SINet	A0	43	NC4K	0.751545	0.088874
SINet	A0	45	NC4K	0.757700	0.090228
SINet	A2	42	NC4K	0.760908	0.088385
SINet	A2	43	NC4K	0.762933	0.087466
SINet	A2	45	NC4K	0.765789	0.090892
SINet	B	42	NC4K	0.764108	0.087897
SINet	B	43	NC4K	0.768972	0.085001
SINet	B	45	NC4K	0.767632	0.088175
SINet	C10	42	NC4K	0.765296	0.086273
SINet	C10	43	NC4K	0.771241	0.085305
SINet	C10	45	NC4K	0.770503	0.086622
SINet	CINV	42	NC4K	0.774697	0.083144
SINet	CINV	43	NC4K	0.774114	0.084030
SINet	CINV	45	NC4K	0.768136	0.084570
SINet	CSHUF	42	NC4K	0.766713	0.085734
SINet	CSHUF	43	NC4K	0.767983	0.085825
SINet	CSHUF	45	NC4K	0.767229	0.089734
SINet	MT	42	NC4K	0.752869	0.108067
SINet	MT	43	NC4K	0.746708	0.101192
SINet	MT	45	NC4K	0.749400	0.104323
SINetv2	A0	42	NC4K	0.745691	0.092348
SINetv2	A0	43	NC4K	0.736067	0.099838